

China Strategies in the Automotive Industry

Governance, Financing, and Long-Term Value Creation

Seminar „M&A as Transformer of the Automotive Industry“

Chair of Banking and Finance — Goethe-Universität Frankfurt

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Introduction

Setting the Stage

55% of all new cars in China are electric in 2025 — who survives?

The market has already tipped

- EV share China 2025: **~55%** of all new registrations (IEA 2026)
- Chinese OEMs: **60%** of all global EV sales
- **70%** of BEVs sold in China cost less than the average car
- IEA forecast: **>90%** EV share in China by 2035

Research Question: Which governance and financing structure creates long-term shareholder value in a technology-intensive, fast-moving market?

Three cases under identical market conditions: **SAIC · BYD · Geely**

Three paths, one market — sharply divergent outcomes

The cases at a glance

Model	State-directed JV	Organic vertical integration	M&A-driven private
Control	Shanghai SASAC (63.71%)	Wang Chuanfu, founder (17%)	Li Shufu, private (100% ZGH)
Net profit 2024	RMB 1.67bn (−88%)	RMB 40.25bn (+34%)	RMB 16.8bn (+240%)
EV Readiness	Medium — JV restructuring ongoing	High — ICE exit 2022	Medium — platform-dependent

Central thesis: Governance structure is the binding constraint across all dimensions — it determines which financing strategies are viable, which capabilities can be developed autonomously, and ultimately which competitive positions remain accessible under rapid technological change.

Theory & Literature Review

Four Theoretical Lenses

Four theoretical lenses — one analytical framework

1. Political Governance & SOE Efficiency

- Shleifer & Vishny (1994): politicians maximize employment, not firm value
- Fan, Wong & Zhang (2007): politically connected CEOs → **-18% CAR**, -21% revenue growth
- Megginson & Netter (2001): mixed-ownership firms no more profitable than pure SOEs

2. JV Dynamics & Capability Transfer

- Holmes, McGrattan & Prescott (2015): JV as implicit technology tax
- Howell (2018): JV rents crowd out autonomous innovation (*"like opium"*)
- Inkpen & Beamish (1997): JVs become unstable when asymmetric learning ends

3. Cross-border M&A & Springboard Strategy

- Luo & Tung (2007): emerging-market firms use acquisitions as springboard
- Zheng, Noorderhaven & Du (2022): 4-stage integration (distancing → diversifying)

4. Dynamic Capabilities & Founder Control

- Teece, Pisano & Shuen (1997): sustainable advantage through path-dependent tacit knowledge
- Zhang et al. (2025): founder control → more breakthrough innovation

Governance

Who Controls the Steering Wheel?

SAIC: State as controller — politically embedded objectives

Ownership structure embeds political goals structurally

- Shanghai SASAC: **63.71%** + COSCO 5.91% + Yuejin 3.60% = **~73% state-owned**
- "Two consistencies" in Annual Report: Party leadership + modern enterprise system
- Party Committee: not informal — formally constitutive element of governance

Empirical consequence (*Fan et al. 2007, n=790 IPO firms*)

Metric	Politically connected	Non-connected
3-year CAR	-18%	Baseline
Revenue growth	-21%	Baseline
Profit growth	-24%	Baseline
Board bureaucrats	36%	19%

2025 Reform: Board of Supervisors abolished → strengthened Audit Committee

→ *Formal modernisation, but 63.71% SASAC stake + Party mandate remain unchanged*

BYD & Geely: Founder control without political constraints

BYD — Consolidated founder control

- Wang Chuanfu: Chairman + CEO since 1995, ~17% direct; with Lu Xiangyang (5.1%) **>1/3 voting rights**
- No Party Committee directive; subsidies (RMB 10.4bn) grant **no control rights**
- Decisions without consensus processes: ICE exit 2022, BYD Semiconductor, FinDreams spin-off
- Wang Chuanfu: *"Fast decision-making"* as one of three central competitive advantages (He 2025)

Geely — Private conglomerate leader

- Li Shufu: 100% owner ZGH (unlisted apex) → **~76% control** over Geely Auto HK
- No JV-partner or state-committee approval needed
- Acquisition strategy: Volvo 2010 (\$1.8bn) → LEVC 2013 → Lotus 2017 (51%) → Smart JV 2020 (50:50 Mercedes)
- **Risk:** multi-brand portfolio generates growing coordination costs

Financing

Capital Structure as Strategic Bet

SAIC & BYD: Capital trap vs. self-financing resilience

SAIC — Capital-light JV model becomes a capital trap

- JV model: VW + GM each bear 50% investment and risk → stable dividends, no own capex
- 2024: SAIC-GM sales **-56.54%** → SAIC must absorb 50% impairment without restructuring control
- Result: Revenue -15.73% (RMB 627.59bn), Net profit **-88.19%** (RMB 1.67bn), ROE **0.58%**
- EU countervailing duty: **35.3%** (vs. BYD 17.0%, Geely 18.8%) — highest subsidy intensity of all three

BYD — Internal financing as structural strength

Metric	FY2024
Revenue	RMB 777.1bn (+29%)
Operating Cash Flow	RMB 133.5bn
Cash position	RMB 154.9bn
R&D spend	RMB 54.2bn (+35.7%)
Interest-bearing debt / total liabilities	4.9%

*In **13 of 14 years** (2011–2024): R&D expenditure exceeded annual net profit (He 2025)*

Geely: Leverage for speed — highest financial risk

Volvo Deal 2010 — The paradigm case

Purchase price	\$1.8bn → total capital ~\$2.7bn
Financing	Bank of China/CCB/ExIm ~\$2.1bn + ~\$932m (~50%) central government
ZGH debt	RMB 16.05bn (2009) → RMB 71.07bn (2010) ; D/C ratio 73.47%
Return	Volvo IPO October 2021 ~ \$18.5bn — 10× purchase price

Current risks (2024/25)

- **Polestar** (Nasdaq PSNY): Net loss \$1.17bn FY2023, Deloitte going-concern warning, only \$103.1m cash end-2024, ZGH as guarantor, ~\$2bn capital need through 2028
- **Lotus Technology**: Cumulative deficit ~\$3.2bn (FY2025), Revenue **-44%** FY2025

Comparison: BYD = growth speed vs. resilience · SAIC = risk hedge vs. loss of control · Geely = balance sheet risk vs. strategic speed

Capability Building

Build, Buy, or Borrow?

SAIC and the Capability Paradox

"JV rents are like opium" — He Guangyan, former Minister of Machinery

What the JV model gave (*Bai et al. 2025, n=all auto models 2001–2014*)

- **+8.3%** quality improvement in JV-affiliated models vs. non-affiliated
- Defect rate: 65% higher in 2001 → **33% higher** in 2014
- Transfer mechanism: worker mobility + supplier networks (*not patent transfer*)

What the JV model prevented (*Howell 2018*)

- JV firms responded to regulatory pressure with quality reduction rather than innovation
- Cannibalization dynamic: own brand competes with profitable JV revenue stream

Capability Paradox: JV transferred production quality but suppressed the autonomous innovation needed for the EV era (*no proprietary battery, no software platform*)

Facility	Capacity	Production	
SAIC-GM	1.452m	534,000	37%
SAIC-VW	1.920m	1.058m	55%
VW MEB (EV)	240,000	34,464	14%
Own-brand (PVB)	—	—	76–95%

BYD & Geely: Organic depth vs. M&A speed

BYD — Organic internal build: structurally hard to imitate

- 1995: mobile phone batteries → 2003: vehicle production → mid-2000s: automotive batteries
- *"Specialised Vertical Integration"*: competencies stack on each other (Wang, Zhao & Ruet 2022)
- Teece et al. (1997): tacit knowledge in routines → **not transferable** via licence or talent recruitment
- BYD did not follow policy agenda — it actively co-designed it (Whitfield & Wuttke 2026)

Geely — M&A as capability compressor

- Volvo 2010: immediate access to safety engineering, 3 platforms, 2,400+ dealer network
- Strategic asset-seeking FDI (Luo & Tung 2007)
- 4-stage integration (Zheng et al. 2022): distancing → balancing → building → **diversifying**
- CMA Platform (Compact Module Architecture): Volvo 40-series, Lynk & Co, Geely — bilateral co-development
- R&D expenditure: **+600%** in 7 years post-acquisition
- Konda et al. (2022): measurable patent convergence → genuine co-creation effect

Long-Term Viability

Who Has a Sustainable Position?

SAIC: JV erosion meets EV transition simultaneously

Double pressure: structural decline in JV, structural weakness in EV

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Structural squeeze

- Domestic: JV demand collapsing, own-brand running at capacity
- Export: EU duty **35.3%** — twice as high as BYD (17.0%)

Response "JV 2.0": Repurposing JV capacity for own-brand EVs; SAIC SHANGJIE with Huawei (Sept. 2025)

***However:** SASAC 63.71% + Party mandate structurally limit speed of capital reallocation*

BYD & Geely: EV transition complete vs. strong core, risky periphery

BYD — EV transition structurally complete

- 2022: full ICE exit — no combustion segment weighing on capital allocation
- FY2025: **>1m exported vehicles** (+140% YoY), 119 countries → reduces home-market dependency
- 4th consecutive year as global NEV market leader
- **USD 5.6bn** capital raise March 2025 (institutional investors + sovereign wealth funds)
- Allen et al. (2024): externally listed private Chinese companies perform at global market level

Structural risks BYD: Gross margin 19% (2024) → 18% (2025); Net profit **-19%** (RMB 32.6bn); net-cash → net-debt via global factory expansion; EU duty 17% (combined 27%)

Geely — Strong core, risky periphery

Entity	Performance
Geely Auto	Revenue +32% (\$33.4bn), Net profit +240%
Volvo Cars	SEK 400.2bn, 6.8% EBIT — second record year
Geely stock	+39% early 2026 vs. BYD +12%
Polestar	Going concern, \$103m cash, ~\$2bn capital need
Lotus Technology	Revenue -44% , cumulative deficit \$3.2bn

Generalizable Lessons

What Can Other Markets Learn?

Lesson 1: Mode of capability acquisition determines durability across technology cycles

	(Mandatory JV)	(Organic)	(M&A)
Short-term effect	+8.3% quality (Bai et al. 2025)	Slow	Immediate platforms
Long-term effect	Autonomous innovation crowded out	Proprietary, non-imitable	Bilateral co-development
EV readiness	Missing	High	Platform-dependent
Cost	Political control	Time	Leverage & complexity

Core finding: Mandatory Technology Transfer creates short-term quality gains but crowds out the autonomous innovation needed for the next technology cycle. Voluntary acquisition and organic development avoid this — at different costs: speed vs. depth vs. independence.

Lessons 2 & 3: Governance and capital structure as first-order constraints

Lesson 2: Governance is a first-order determinant of strategic adaptability

In fast technology markets, adaptation speed is itself a competitive advantage. That speed is not a resource question — it is a question of **control-rights distribution**.

	Decision speed	Constraint
SAIC	Slow	SASAC + Party Committee + 50:50 JV boards
BYD	Fast	Consolidated founder control
Geely	Medium	Private apex, but multi-brand coordination costs

"When the rate of technological change exceeds the decision cycle time of a governance structure, governance itself becomes the binding competitive constraint."

Lesson 3: Capital structure determines resilience when markets contract

	Structure	Strength	Vulnerability
SAIC	Capital-light	Risk hedge	50% impairment without restructuring control
BYD	Self-financing	Resilient to partner losses	Margin compression under price war
Geely	Leverage for speed	10× Volvo return	Portfolio liabilities (Polestar, Lotus)

Capital structure is not a static optimisation — it is a dynamic bet on the firm's ability to retain operational control over its primary risk exposure.

Conclusions

Governance as Binding Constraint

Governance is the binding constraint across all four dimensions

Central finding of this paper

Governance structure determines which financing strategies are viable, which capabilities can be developed autonomously, and which competitive positions remain accessible under rapid technological change.

Model assessment

	Governance	Financing	Capabilities	EV Readiness	
BYD	Founder control ✓	Self-financing ✓	Organic, deep ✓	Complete ✓	Structurally superior
Geely	Private apex ✓	Leverage ⚠	M&A-acquired ✓	Platform-dependent ⚠	Strong core, risky periphery
SAIC	State/Party ✗	Capital trap ✗	JV-dependent ✗	Transitioning ✗	Double structural brake

Open question: Can SAIC's "JV 2.0" overcome the political-bureaucratic decision speed that has historically limited its adaptability?

AI Highlights — Shared Git Repository + Claude (Anthropic)

What we used

- Shared git repository with three separate company wikis (BYD, SAIC, Geely)
- Claude Code CLI with `wiki-ingest` , `wiki-query` , `wiki-lint` agents
- **29+ academic sources** structured, ingested, cross-referenced
- Source-rating system for source prioritisation

What worked well

- **Wiki-Ingest:** Automatic extraction and linking of entities and concepts across sources
- **Cross-source synthesis:** Contradictions between sources automatically flagged (e.g. ROE discrepancy)
- **Writing assistance:** Sections drafted directly from cited sources

Challenges

- No direct SharePoint/cloud access → files had to be stored locally
- Hallucinations on financial figures → consistent cross-checks against annual reports required
- Wiki schema (frontmatter, linking, index/log) required initial setup (~1–2h)

Cost: Claude Pro ~\$20/month